moderate rainfall. In the same way cold front is formed due to advent of cold air mass which leads to heavy rainfall with lightning and thunder.

6.6.3 Cloud Burst

A 'cloud burst' is a sudden aggressive rainstorm falling in a short period of time limited to a small geographical area. Meteorologists say that the rain from a cloud burst is usually of the heavier rain with a fall rate equal to or greater than 100 mm (3.94 inches) per hour. Generally cloudbursts are associated with thunderstorms. The air currents rushing up words in a rain storm hold up a large amount of water. For example cloud bursts in the region of Uttarkhand (2013) and Chennai (2015).

6.7 Atmospheric Disturbances (Cyclone and Anti Cyclone)

The atmospheric disturbances which involve a closed circulation of air around a low pressure at centre and high pressure at periphery, rotating anticlockwise in northern hemisphere and clockwise in southern hemisphere is called 'Cyclones' (Figure 6.28). Cyclones may be classified into two types based on latitude of its origin.

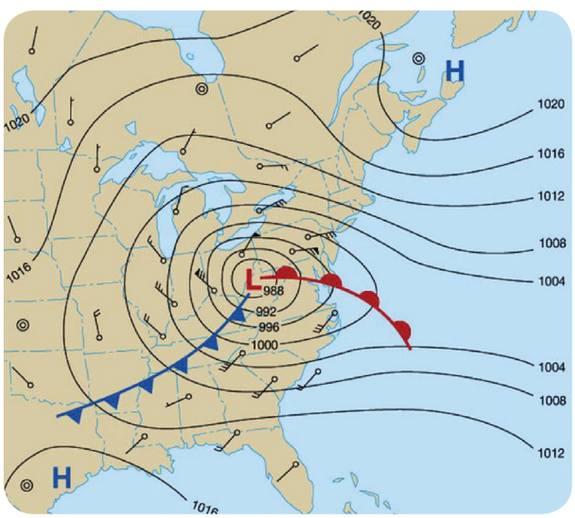


Figure 6.28 Cyclone

They are:

A. Tropical cyclone B. Temperate cyclone.

A. Tropical Cyclone

Cyclone formed in the low latitudes is called as Tropical cyclone. They form over warm ocean waters in the tropical regions. The warm air rises, and causes an area of low air pressure.

6.7.1 Stages of Development of Tropical Cyclone

As per the criteria adopted by the World Meteorological Organisation (W.M.O.), India Meteorological Department classifies the low pressure systems in to vary classes based on wind speed.

1. Tropical Disturbances
2. Tropical depressions Low winds with a speed between 31 and 61 km ph.
3. Tropical cyclone wind speed from 62 to 88 km ph and it is assigned a name.
4. Severe Cyclonic Storm (SCS) wind speed is between 89 to 118 km ph
5. Very SCS wind speed between 119 to 221 km ph and
6. Super Cyclonic Storm when wind exceeds 221 km ph.

6.7.2 Origin of Tropical Cyclone

Tropical cyclones have certain mechanism for their formation. These are

A source of warm, moist air derived from tropical oceans with sea surface temperature normally near to or in excess of 27 °C (Figure 6.29).

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